Example 1:
Suppose that each letter represents a state in a solution space. The following moves are legal :

A5 to B4 and C4
B4 to D6 and E5
C4 to F4 and G5
D6 to I7 and J8
I7 to K7 and L8

Start state : A
Goal state : E

The numbers next to the letters represent the heuristic value associated with that particular state.

	OPEN	**CLOSED**	**X**	**X's Children**	**State**
1	A5	-			failure
2			A5	B4, C4	failure
3	B4, C4	A5			failure
4	C4	A5	B5	D6, E5	failure
5	C4, E5, D6	A5, B4			failure
6	E5, D6	A5, B4	C4	F4, G5	failure
7	F4, E5, G5, D6	A5, B4, C4			failure
8	E5, G5, D6	A,5 B4, C4	F4	none	failure
9	E5, G5, D6	A5, B4, C4, F4			failure
10	G5, D6	A5, B4, C4, F4	E5		success

11. Hill-Climbing

Hill-climbing is similar to the best first search. While the best first search considers states globally, hill-climbing considers only local states. The application of the hill-climbing algorithm to a tree that has been generated prior to the search is illustrated in **Figure 11.1**.

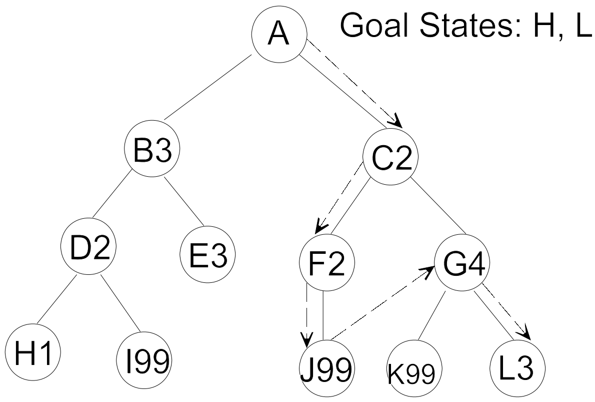

Figure 11.1

The hill-climbing algorithm is described below. The hill-climbing algorithm generates a partial tree/graph.

Hill Climbing Algorithm

```
def hill_climbing(in Start, out State)
 open = [Start], close = [],
 State = failure;
 while (open <> []) AND (State <> success)
   begin
    remove the leftmost state from open, call it X;
    if X is the goal, then
      State = success
    else begin
      generate children of X;
      for each child of X do
      case
        the child is not on open or closed
         begin
          assign the child a heuristic value,
         end;
        the child is already on open
          if the child was reached by a shorter path then
           give the state on open the shorter path
        the child is already on closed:
         if the child is reached by a shorter path then
          begin
           remove the state from closed,
         end;
       endcase
     endfor
     put X on closed;
     re-order the children states by heuristic merit (best leftmost);
     place the reordered list on the leftmost side of open;
   endwhile;
   return State;
end.
```

Example 1: Suppose that each of the letters represent a state in a state space representation. Legals moves:

A to B3 and C2
B3 to D2 and E3
C2 to F2 and G4
D2 to H1 and I99
F2 to J99
G4 to K99 and L3

Start state: A
Goal state: H, L

The result of applying the hill-climbing algorithm to this problem is illustrated in the talbe below.

	OPEN	**CLOSED**	**X**	**X's Children**	**State**
1	A	-			failure
2			A	B3, C2	failure
3	C2, B3	A			failure
4	B3	A	C2	F2, G4	failure
5	F2, G4, B3	A, C2			failure
6	G4, B3	A, C2	F2	J99	failure
7	J99, G4, B3	A, C2, F2			failure
8	G4, B3	A, C2, F2	J99	none	failure
9	G4, B3	A, C2, F2, J99			failure
10	B3	A, C2, F2, J99	G4	K99, L3	failure
11	L3,K99, B3	A, C2, F2,J99, G4			failure
12	K99, B3	A, C2, F2, J99, G4	L3		success